

National University of Computer and Emerging Sciences, Lahore Campus  
**Quiz .....2 [BS(CY): Section 3A] Fall 2026**

**Cyber Security (CY2004)**

**Quiz Date: September 15, 2026**

**Total Marks: 20**

**Duration: 20 -Minutes**

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Name ----- Roll #----- Section -----

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Provide precise answers to the following questions.

**Question 1:** A stream cipher uses a keystream generator to encrypt a plaintext message. The encryption formula is  $C = P \oplus K$ . Given the following 8-bit Plaintext (P) and Seed Key (K), calculate the Ciphertext (C). Then, verify your answer by decrypting the Ciphertext back to Plaintext.

**Plaintext (P):** 1 0 1 1 0 0 1 0

**Keystream (K):** 0 1 0 1 1 0 1 0

**Question 2:** A symmetric block cipher processes data in fixed-size blocks. Suppose you are using a block cipher that operates on a block size of 128 bits. You need to encrypt a plaintext file that is exactly 500 bytes in size.

- How many bits is the entire plaintext file?
- How many full 128-bit blocks can be formed?
- If there are remaining bits, how many are there? (Assume the cipher requires padding for the final block).

**Question 3:** Encrypt "WELCOME" with a random key using A=1...Z=26

You are provided with the following random Key for each letter of the message:

W → Z;          E → M;          L → Q;          C → X;  
O → N;          M → F;          E → T

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**Answer to Question 1:**

Encryption ( $C = P \oplus K$ ):

- $1 \oplus 0 = 1$
- $0 \oplus 1 = 1$
- $1 \oplus 0 = 1$
- $1 \oplus 1 = 0$
- $0 \oplus 1 = 1$
- $0 \oplus 0 = 0$
- $1 \oplus 1 = 0$
- $0 \oplus 0 = 0$

**So Ciphertext (C):** 1 1 1 0 1 0 0 0

**Verification:** Decryption ( $P = C \oplus K$ )

- $1 \oplus 0 = 1$
- $1 \oplus 1 = 0$
- $1 \oplus 0 = 1$
- $0 \oplus 1 = 1$
- $1 \oplus 1 = 0$
- $0 \oplus 0 = 0$
- $0 \oplus 1 = 1$
- $0 \oplus 0 = 0$

**Decrypted Plaintext:** 1 0 1 1 0 0 1 0 (Matches original)

**Answer to Question 2:**

- Total bits:  $500 \text{ bytes} \times 8 \text{ bits/byte} = 4000 \text{ bits}$
- Full blocks:  $4000 \text{ bits} \div 128 \text{ bits/block} = 31.25$ .  
So, there are 31 full blocks.
- Remaining bits:  $4000 - (31 \times 128) = 4000 - 3968 = 32 \text{ bits}$ .  
(This means the 32nd block will only have 32 bits of data and will require 96 bits of padding to fill the 128-bit block size).

**Answer to Question 3:**

**Step 1:** Convert Plaintext and Key to Numbers (A=1, Z=26)

| Plain Text | Number (Value) | Corresponding Key | Number (Value) |
|------------|----------------|-------------------|----------------|
| W          | 23             | Z                 | 26             |
| E          | 5              | M                 | 13             |
| L          | 12             | Q                 | 17             |
| C          | 3              | X                 | 24             |
| O          | 15             | N                 | 14             |
| M          | 13             | F                 | 6              |
| E          | 5              | T                 | 20             |

**Step 2: Calculate the Ciphertext Numbers**

Apply the formula:  $C = (P + K) \pmod{26}$

**Note :** If the result is greater than 26, subtract 26 to get the remainder.

- ✓  $W (23) + Z (26) = 49. 49 \pmod{26} = 23$  (Since  $49 - 26 = 23$ )
- ✓  $E (5) + M (13) = 18. 18 \pmod{26} = 18$
- ✓  $L (12) + Q (17) = 29. 29 \pmod{26} = 3$  (Since  $29 - 26 = 3$ )
- ✓  $C (3) + X (24) = 27. 27 \pmod{26} = 1$  (Since  $27 - 26 = 1$ )
- ✓  $O (15) + N (14) = 29. 29 \pmod{26} = 3$  (Since  $29 - 26 = 3$ )
- ✓  $M (13) + F (6) = 19. 19 \pmod{26} = 19$
- ✓  $E (5) + T (20) = 25. 25 \pmod{26} = 25$

**Step 3: Convert Numbers back to Letters**

Using the A=1, B=2... Z=26 mapping:

- ✓  $23 = W$
- ✓  $18 = R$
- ✓  $3 = C$
- ✓  $1 = A$
- ✓  $3 = C$
- ✓  $19 = S$
- ✓  $25 = Y$

Thus, finally, The encrypted ciphertext for "WELCOME" is **WRCACSY**.